

Explainable AI for academic integrity verification

Supervisor meeting, 4 August 2026

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Submission: end of August 2026. Four weeks remain.

Every component is built, evaluated and written up

12 / 12

chapters drafted

26,700

words in the body

41

references, each verified

32

figures, all cross-referenced

The pipeline runs end to end on the laptop: detect, explain, extract claims, generate questions, assemble the guide.

The consistency audit (numbers against result files, citations both ways, figure numbering) runs clean before every build.

Remaining work is writing and revision, not new components. That was the plan for weeks 11 to 13 and we are on it.

SINCE WE LAST MET

What moved

The evaluation is now finished. The three-judge panel completed at sixty questions, and the multi-generator test finished at forty essays per generator.

The v4 question experiment closed out. Questions built never to name the claim's content reach 0.27 against the 0.30 generic ceiling, and with the gate applied at generation time they are 100 percent content-free at statistically the same score.

An examiner-lens audit of the whole document found formal defects that would have cost marks. All fixed, described on the next slide.

A full visual demonstration is built: an interactive walkthrough on a real essay pair, plus an eighteen-scene auto-playing film that explains the system with no narration.

The repository is current on GitHub, private, with no data, models, keys or working notes in its history.

The document was failing the ATU template in eight ways

Found

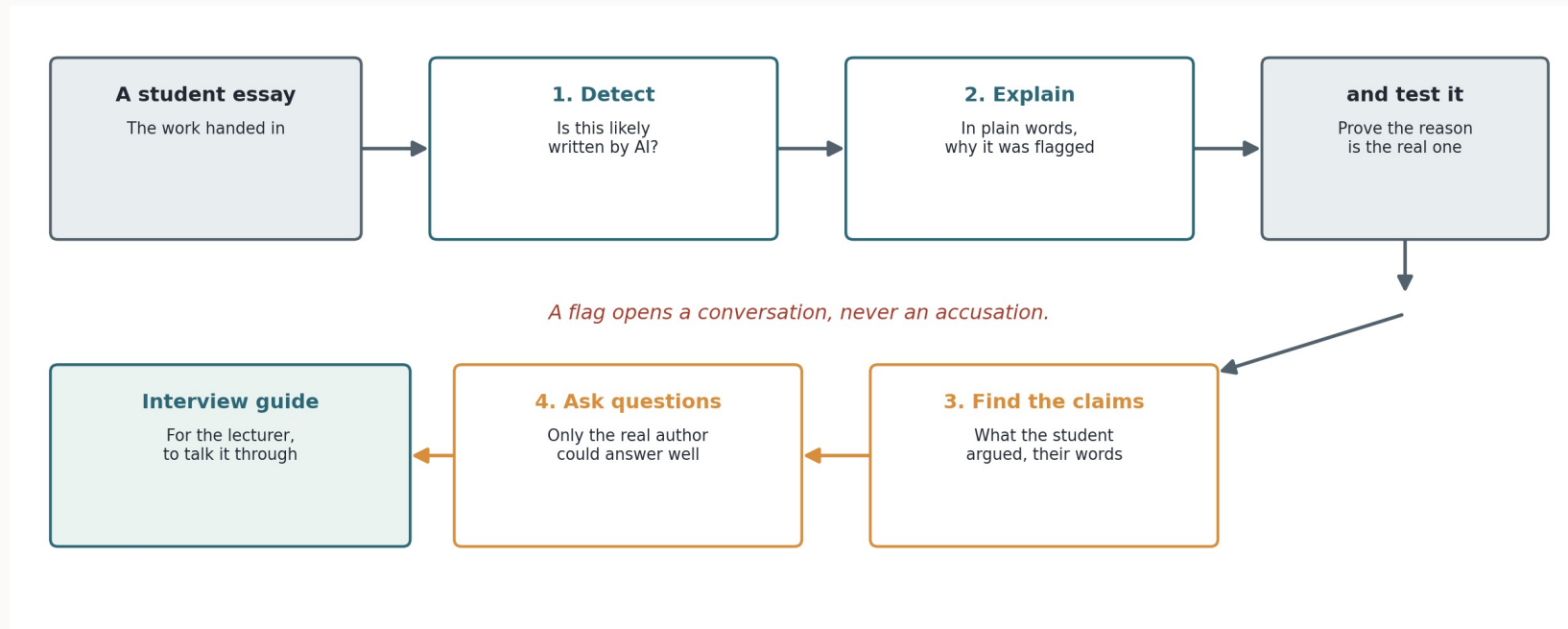
- Abstract 482 words against a hard 250-word cap
- Appendix tables rendered as raw pipe characters
- Code listing collapsed, all indentation lost
- Table of Figures three figures out of date
- No Table of Tables, no Table of Code Listings
- Figures numbered out of order in two chapters
- One cross-reference pointed at the wrong figure
- Six figures never referred to in the body text

Fixed

- Abstract rewritten at 250 words, results intact
- Real Word tables, captioned and numbered
- Listings keep indentation, captioned
- Front matter is generated from the chapters now
- Both required tables produced automatically
- Renumbered into appearance order, refs moved with them
- Corrected, and the audit now checks ordering
- Each figure now has a body-text reference

Three new checks were added to the audit script so none of these can come back before hand-in.

One pipeline, five stages, all on the laptop



Submission in, Verification Interview Guide out. Every stage runs locally, so student work never leaves the institution.

What a lecturer receives

Verification Interview Guide

Submission: 3108a | **Generated:** 2026-07-11 | **Question backend:** local-hf:v3:qg_finetune_qwen3b_v3

This guide is evidence for a conversation with the student, not an accusation. The detector's judgement is fallible, and the fair use of this document is to ask, listen, and weigh the answers.

1. Detection summary

The hybrid detector (transformer fused with stylometric features and GPT-2 perplexity) scores this submission at **0.9572** probability of being AI-generated, so it is **flagged as likely AI-generated**. No score of 1.0 exists on this scale: the model is never certain, only confident. On matched in-domain test data the detector's F1 is 0.99; on out-of-domain academic text its false-positive rate rises sharply, so treat the score as a reason to talk, never as proof.

Component views: transformer 0.9996, style-plus-perplexity 0.9985. The hybrid fuses both because the style half keeps the detector calmer on unusual but human writing.

Why this particular score? The habits that moved it, against typical student writing:

- A language model finds this text unusually easy to predict (surprise score 18 against a typical 30); machine text tends to be predictable, which here points toward AI.
- Fewer helper verbs (is, has, can) than typical (0.036 against 0.053), which here points toward AI.
- The overall word variety is lower than typical (9.3 against 14.0), which here points toward AI.
- The sentences are unusually uniform in length (variation 7.5 against a typical 11.4), which here points toward AI.
- Fewer one-off, unusual words than typical (0.08 against 0.18), which here points toward AI.

The guide, generated end to end with live models:

- The detection verdict, with both readers shown separately
- The explanation card: five named writing habits against typical student writing
- The student's own claims, with the sentence numbers they came from
- Verification questions per claim, each tagged with the level of thinking it asks for
- A first page that frames all of it as evidence for a conversation, not a verdict

Headline results, every one traced to a result file

F1 0.99

detection in domain, on the audited corpus

100%

of essays caught from two unseen generators

24 of 29

essays where the laptop model beat the commercial tier

79% to 61%

false accusations cut by fusing in writing style

0 of 3

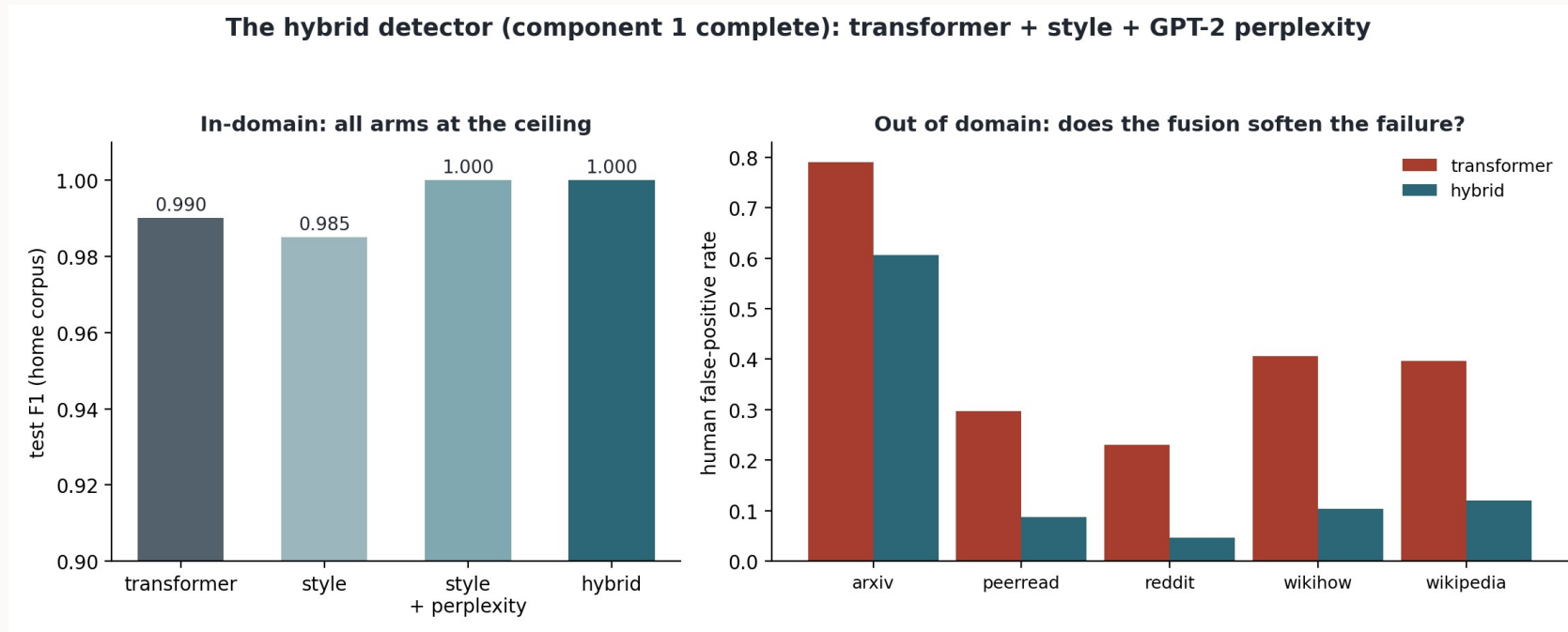
AI judges that track the objective measure

3

of my own headline numbers retracted by my own checks

The teal results are what the system achieves. The rust results are the honest limits, and they shape the design.

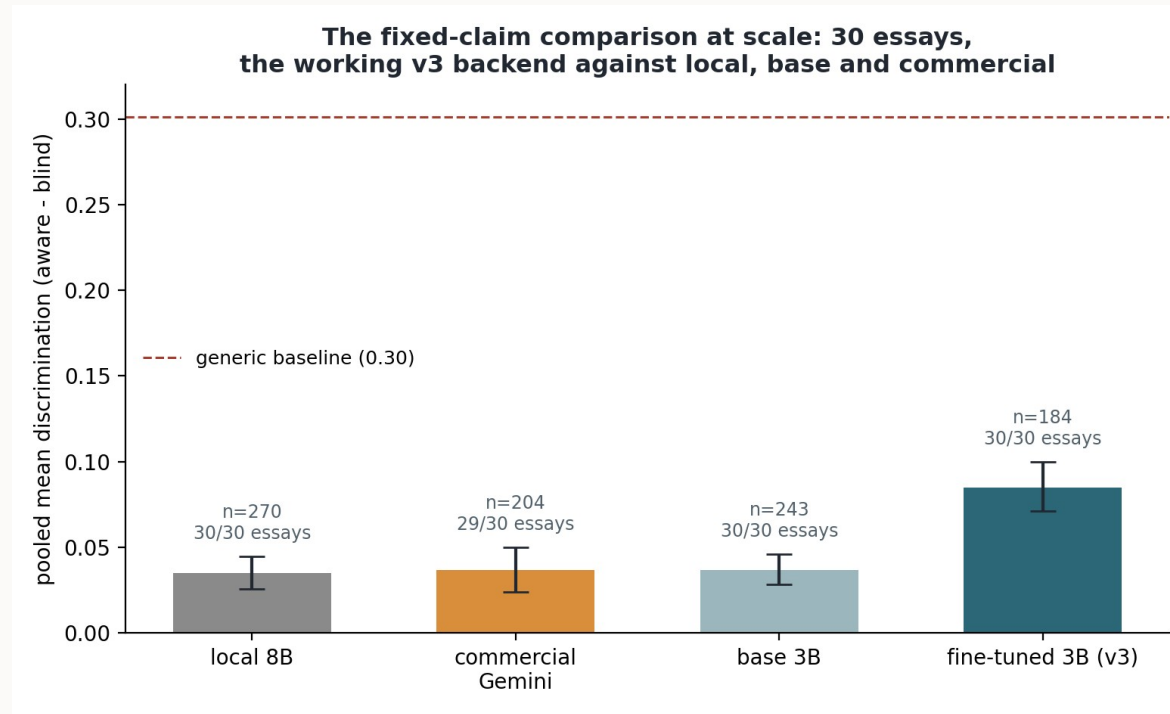
The failure that matters is the false accusation



In domain every arm is at the ceiling. Out of domain the fusion barely moves F1 but cuts wrongful flags of human writing three to eight times.

THE STRONGEST RESULT

The laptop model beat the commercial tier



Thirty essays, 901 questions, identical claims for every writer. v3 beats the free commercial tier on 24 of the 29 shared essays ($p = 0.0003$), with every question well-formed.

Two decisions, and both are time-critical

1. The validation study: go or no go?

The protocol is drafted and includes a blind arm that would settle the v3 versus v4 question on real people. With four weeks to submission, my honest view is that it cannot be run, analysed and written up in time, and that ethics approval alone would consume much of it. I propose we keep it as the primary piece of future work, clearly framed in Chapter 10, rather than start it now and risk the write-up.

2. Which question backend ships: v3 or v4?

v3 asks specific, quotable questions. v4 never names the claim's content, so it is harder to fake but more generic in phrasing, and it scores higher. Nothing technical blocks either. Without the human study to decide it, I lean to shipping v3 and reporting v4 as the measured alternative, because v3 is what a lecturer can actually read aloud in an interview. I would like your view.

What could still go wrong, and what I am doing about it

Time. Four weeks, and the remaining work is all writing.

Writing weeks are protected. If anything slips I cut analysis depth, not the write-up.

The literature review is thin at 2,000 words for nine themes.

Expanding the five sections that carry the biggest contributions, using sources already verified.

Results are not yet positioned against published baselines.

Adding reported figures from sources already in the reference list, so my numbers can be judged.

No human validation of the questions.

Stated as the central limitation and the first piece of future work. The simulation is a conservative floor.

Plan to submission

Week of 4 August

Literature review expansion and published baselines into the results chapters.

Week of 11 August

Full revision pass over Chapters 1 to 8, methodology account strengthened.

Week of 18 August

Discussion and conclusions finalised, appendices completed, figures regenerated.

Week of 25 August

Proofreading, template check, print check, submission with days in hand.

IN ONE LINE

The system works, the document is close, and the risks are the ones I can name

I would value your steer on the two decisions, and any early sight of what you want strengthened in the write-up.

Happy to send the current draft of Chapters 1 to 8 straight after this meeting.